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The role of analytical reasoning and source credibility on the evaluation of real and fake full-length news articles

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Abstract:
Aim: *Previous research has focused on accuracy associated with real and fake news presented in the form of news headlines only, which does not capture the rich context news is frequently encountered in real life. Additionally, while previous studies on evaluation of real and fake news have mostly focused on characteristics of the evaluator (i.e., analytical reasoning), characteristics of the news stimuli (i.e., news source credibility) and the interplay between the two have been largely ignored. To address these research gaps, this project examined the role of analytical reasoning and news source credibility on evaluation of real and fake full-length news story articles.*

Method: *We conducted two independent but parallel studies, with Study 2 as a direct replication of Study 1, employing the same design but in a larger sample (Study 1: N = 292 vs. Study 2: N = 357). In both studies, participants viewed 12 full-length news articles (6 real, 6 fake), followed by prompts to evaluate each article's veracity and credibility.*

Findings & Conclusions: *Consistent across both studies, higher analytical reasoning was associated with greater fake news accuracy and lower perceived credibility for fake news, but not for real news. Lower analytical reasoning was associated with greater accuracy for real news from credible sources. Cues like news source credibility systematically affect evaluation.*

Introduction

Fake news refers to 'fabricated information that mimics news media content in form but not in organizational process or intent' (Lazer et al., 2018, p. 1094). Broad use of online media platforms in modern society has drastically increased access to news but also increased distribution of misinformation. Recent polls indicate that a significant portion of Americans (47%) report having difficulty distinguishing between real and fake news (Associated Press, 2019).

A Cognitive Account of Fake and Real News Detection

According to Dual-Process Theory, individuals engage in two modes of information processing: a quick, intuitive mode (System 1) and a slow, deliberate mode (System 2; De Neys, 2012; Kahneman, 2011; Stanovich, 2009). System 1 is associated with low analytical reasoning and reliance on cognitive heuristics when making decisions. System 2, in contrast, is associated with high analytical reasoning and involves careful and systematic consideration of information, and therefore is less error-prone.

In line with Dual-Process Theory, individuals who scored higher on a measure of analytical reasoning (i.e., Cognitive Reflection Test [CRT]; Frederick, 2005) were better at detecting fake news than individuals who scored low on analytical reasoning, regardless of whether the news content aligned with their political beliefs (Pennycook & Rand, 2019). Furthermore, engagement in analytic reasoning accounted for 56% to 95% of the variance in accurate detection of fake news.

Research Hypotheses

- **Hypothesis 1 (Veracity):** Higher analytical reasoning will be associated with increased fake news accuracy, while there will be no relationship between analytical reasoning ability and real news detection accuracy.
- **Hypothesis 2 (Moderation):** Lower analytical reasoning will be associated with increased accuracy for real and fake news paired with a credible compared to a non-credible news source.
- **Hypothesis 3 (Credibility):** Higher analytical reasoning will be associated with less perceived credibility for fake news, while analytical reasoning ability will not affect perceived credibility of real news.
- **Hypothesis 4:** Lower analytical reasoning will be associated with greater perceived credibility for real and fake news paired with a credible compared to a non-credible news source.

Methodology

We conducted two parallel but independent studies. Study 1 final sample: N = 292 undergraduate students. Study 2 final sample: N = 357 undergraduate students.

Design: Both studies adopted a 2 (Veracity: real vs. fake; within-subjects) x 2 (Source: credible vs. non-credible; between-subjects) mixed design. Pairs were controlled using NY Times, Washington Post, and NPR (credible sources) and True Pundit, Red State, and Conservative Daily News (non-credible sources) pilot-tested previously.

Cognitive Reflection Test (CRT): The CRT was used to measure analytical reasoning based on 3 trick questions (e.g. the bat and ball problem). Higher score represents higher analytical processing.

Findings & Results

Accuracy and Analytical Reasoning (Hypothesis 1)

Consistent across both studies, the Veracity x CRT interaction on accuracy was significant (Study 1: $p < 0.001$; Study 2: $p = 0.001$). Real news accuracy did not change across levels of analytical reasoning (indexed by CRT scores). Accuracy for fake news, however, increased with higher analytical reasoning. Furthermore, higher analytical reasoning was associated with better detection of fake than real news, supporting the idea that deep cognitive reflection enables spotting visual or logical inconsistencies in fake content.

Source Credibility Effects (Hypothesis 2)

The three-way interaction between Veracity x CRT x Source was significant in Study 2 ($p = 0.013$). Lower analytical reasoning was associated with greater accuracy for real news from credible compared to non-credible sources. However, source credibility did not affect accuracy for fake news across levels of analytical reasoning, indicating that analytical thinkers evaluate the content directly rather than relying on source cues.

Perceived Credibility vs. Accuracy (Hypothesis 3 & 4)

Consistent across both studies, the Veracity x CRT interaction on perceived credibility was significant. Perceived credibility for real news was overall higher than perceived credibility for fake news and was not influenced by levels of analytical reasoning. In contrast, higher analytical reasoning was associated with less perceived credibility for fake news. The credibility of the news source did not affect perceived credibility of fake news in individuals high on analytical reasoning.

Conclusions & Discussion

This study is the first to demonstrate a positive association between analytical reasoning and fake news detection accuracy using full-length news articles, representing a more ecologically valid approach in research on news evaluation. The findings suggest that naturalistic decision-making framework can describe how people evaluate real news automatically based on familiarity, whereas fake news requires effortful System 2 reasoning to spot deception markers. Educating citizens on analytical reflection remains a critical path toward building cognitive resistance against fake news.

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